Application Note

Version history

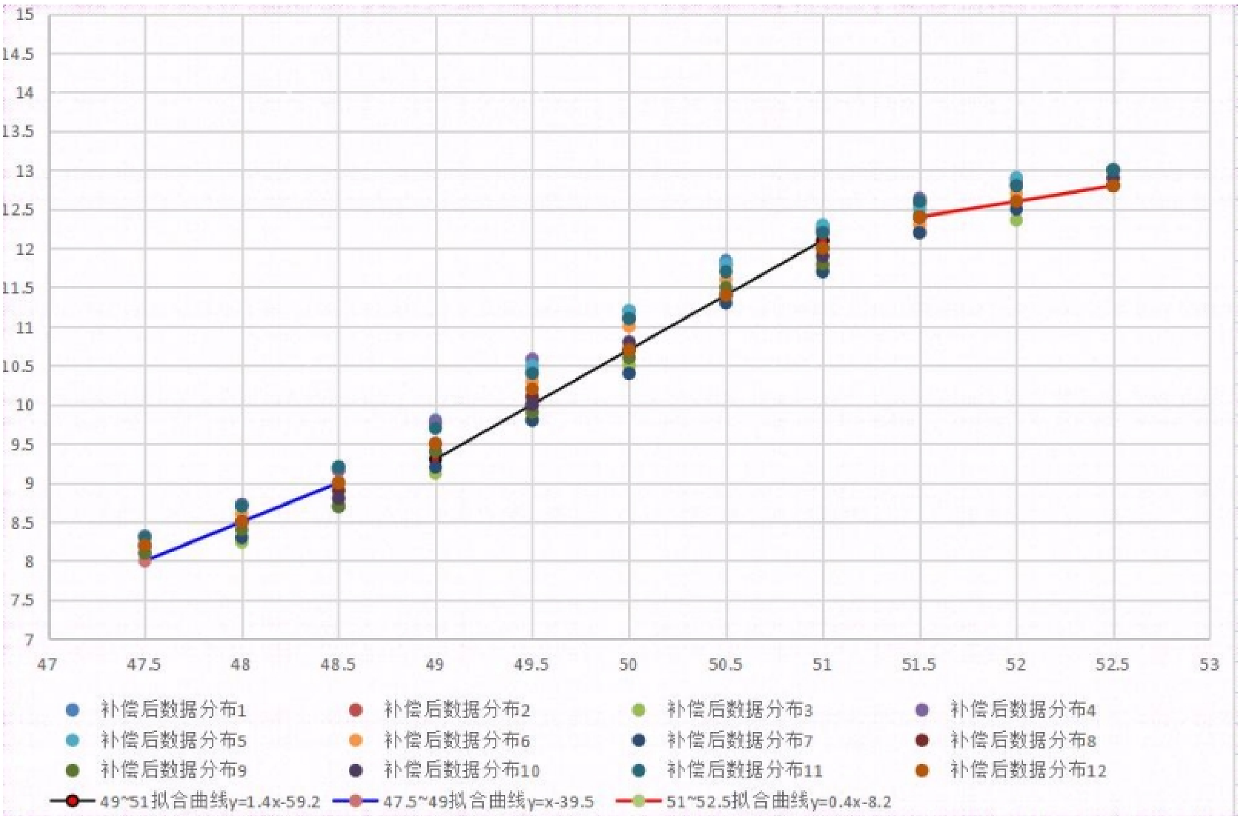
Version	Author	Participator	Release Date	Notes
1.0	Su xiang		2022-8-12	

Content

1 The zero crossing output of V9260S is delayed

The customer reported that the V9260S zero crossing output had a delay, which affected the application judgment.

The following is the RC compensation data distribution diagram and fitting curve of the 12 meters. See Appendix A for detailed data.



2 Cause analysis and solution

2.1 Cause analysis

1、Due to the third-order CIC filtering inside the chip, the frequency of 6400Hz (default configuration) of the fundamental voltage signal processed by the low-pass filter and band-pass filter is sampled, and the zero crossing of the sampling point is determined. The filtering delay is the main reason;

2、A high-frequency RC oscillator is built into the metering chip. RCH error will affect the clock, and the delay will also have errors. Since the compensation step of the chip is 2%, there is still a small amount of RC error even after RC compensation, which is also the reason for the inconsistent delay after compensation;

3、The change of power grid frequency will affect the spectrum distribution and the delay will change;

4、Because there are 128 sampling points in each cycle, there will be 0.156ms (20ms * 1/128) deviation during zero-crossing sampling.

Based on the above four reasons, there will be a certain delay and difference between the zero-crossing output chips.

2.2 Solution

1、According to the deviation of RCH, compensate RCH. The compensation can be initialized and compensated once or at a regular time in the program, and the customer will compensate according to the actual situation.

2、After compensating the RCH, the compensation function is divided into three segments according to the current grid frequency value.

2.3 Compensation method

1、The RCH clock offset rate is obtained：

E=(K * SYS_BAUDCNT8)/(8 * 6553600)

E: RC clock offset, compared with 1

K: MCU actual baud rate, such as: 2400,4800

The calculated RC clock offset rate can be used to calibrate the RC clock register, which is SysCtrl, Bit[11:7].

The reference code is as follows:

```
if(MeasureDrv_ReadRaccoon(T8BAUD,1) == true)
    {
        memcpy((uint8_t *)&uT8BAUD_value, gs_RacCtrl.ucBuf+3, 4);
    }
 E = (float)(4800*uT8BAUD_value)/(8*3276800);
```

After the calculated value is subtracted by one, the error percentage is obtained and compensated according to the instructions in manual SysCtrl Bit[11:7].

Note: If the calculated error percentage is more than 3%, it is better to compensate according to 4%; if the error percentage is more than 2%, it is better to compensate according to 2%. The checksum is calculated after compensation. If the checksum register is written incorrectly, the zero crossing square wave will not occur.

Reference Manual for Checksum Writing Method

2.4 Piecewise Function

According to the 9260S zero crossing output delay after RC compensation, we give three first-order functions as follows:

1、47.5~49HZ：y=x-39.5

2、49~51HZ： y=1.4x-59.2

3、51~52.5HZ：y=0.4x-8.2

Where y is the delay time (ms); X is the frequency, (HZ)

Within each frequency range, the frequency x is substituted into the fitting curve, and the error of the calculated delay time y is within ± 0.5ms

The waveform data captured in this application scheme is the delay time from zero crossing of negative to positive sine wave to square wave output.

Appendix A

Frequency unit: Hz, Time unit: ms

Freq	1#	2#	3#	4#	5#	6#	7#	8#	9#	10#	11#	12#
47.5	8.32	8.31	8.18	8.31	8.3	8.2	8.1	8.2	8.1	8.2	8.3	8.2
48	8.72	8.72	8.24	8.72	8.7	8.6	8.3	8.5	8.4	8.5	8.7	8.5
48.5	9.2	9.16	8.7	9.17	9.2	9	8.7	8.9	8.7	8.8	9.2	9
49	9.8	9.76	9.12	9.77	9.7	9.5	9.2	9.4	9.4	9.5	9.7	9.5
49.5	10.44	10.4	9.8	10.58	10.5	10.3	9.8	10.1	9.9	10	10.4	10.2
50	11.12	11.1	10.5	11.2	11.2	11	10.4	10.7	10.6	10.8	11.1	10.7
50.5	11.84	11.78	11.4	11.8	11.8	11.6	11.3	11.4	11.5	11.4	11.7	11.4
51	12.24	12.2	11.76	12.26	12.3	12	11.7	12	11.8	11.9	12.2	12
51.5	12.6	12.56	12.2	12.64	12.5	12.3	12.2	12.4	12.4	12.4	12.6	12.4
52	12.88	12.84	12.36	12.9	12.9	12.7	12.5	12.6	12.8	12.8	12.8	12.6
52.5	13	13	12.8	13	13	12.9	12.8	12.8	13	12.9	13	12.8